

8 September 2026

Dear Editor-in-Chief,

I am submitting “Future-Context Requirements for Streaming Accent Conversion: A Controlled Study Using Causal Phone Translation” for consideration as a Regular Paper in the IEEE/ACM Transactions on Audio, Speech, and Language Processing.

Streaming foreign accent conversion systems choose a lookahead budget, yet prior work does not densely characterise how that choice trades future context for latency and accuracy. This manuscript studies that relationship using a causal phone translator as a controlled probe rather than claiming an end-to-end accent-conversion evaluation.

The study evaluates 21 lookahead budgets and finds a strong but encoder-dependent future-context response. For the WavLM-based probe, phone error rate decreases approximately 0.045 per doubling of lookahead, with no identifiable knee and a repeatability-aware saturation onset near 340 ms. A pre-specified manner-class hypothesis is not supported, while canonical-phone prediction benefits more strongly from future context than realized-phone transcription. A matched wav2vec 2.0 experiment shows that neither the exchange rate nor the detailed response curve transfers across encoders. The manuscript also separates algorithmic from computational latency and documents measurement failures that materially affected earlier analyses.

The manuscript is nine double-column pages and is submitted as a Regular Paper. Code, configurations, analysis scripts, causality tests and per-condition outputs are publicly available at the repository identified in the manuscript.

A preprint has been submitted to arXiv and is awaiting announcement. The work has not been published elsewhere and is not under consideration by another journal.

Thank you for your consideration.

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